API Specification:

Here is the complete API Specification Document designed specifically for the SALESTORM Flash Sale Platform. It covers high-concurrency stock reservation, order checkout, payment execution, and cancellation flows.

Save this deliverable inside `05_API_Specs/openapi.yaml` or `05_API_Specs/README.md`.

SALESTORM API Specifications (OpenAPI 3.0 Standard)

1. Core API Strategy & Standards
   - Protocol: HTTPS / TLS 1.3
   - Base URL: https://api.salestorm.io/v1
   - Authentication: Bearer <JWT_TOKEN> in Authorization Header
   - Mandatory Headers for Mutations: Idempotency-Key: <UUID_V4> (Prevents duplicate executions on retry)
   - Error Response Format: RFC 7807 Problem Details Standard (application/problem+json)

2. API Endpoint Matrix

Method	Endpoint	Purpose	Key Headers / Params	Success Status
GET	/catalog/flash-sales/{id}	Fetch flash sale items & stock	Standard Headers	200 OK
POST	/reservations	Reserve stock with 10-min TTL	Idempotency-Key	201 Created
DELETE	/reservations/{id}	Cancel reservation & release stock	Standard Headers	200 OK
POST	/orders/checkout	Finalize order from reservation	Idempotency-Key	201 Created

POST	/payments/intent	Initiate idempotent gateway payment	Idempotency-Key	200 OK
POST	/payments/webhook	Gateway asynchronous callback	Gateway Signature Header	200 OK

3. Detailed Endpoint Contracts

3.1 Reserve Inventory (Flash Sale High-Concurrency Endpoint)

Atomically decrements stock using Redis Lua scripts and creates a 10-minute hold.

- HTTP Method: POST
- Path: /reservations
- Headers:
    - Authorization: Bearer <JWT_TOKEN>
    - Idempotency-Key: 7b89d9e4-84c2-4a0b-9311-6b2a0c4f8012

Request Body (application/json)

JSON

```
{
  "product_id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
  "quantity": 1
}
```

Success Response (201 Created)

```
{
  "status": "SUCCESS",
  "data": {
    "reservation_id": "c3a1b2c3-d4e5-4f6a-8b9c-0d1e2f3a4b5c",
    "product_id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
    "quantity": 1,
    "status": "PENDING",
    "ttl_seconds": 600,
    "expires_at": "2026-10-05T12:33:56Z"
  }
```

Error Responses

- 409 Conflict (Out of Stock / Flash Sale Sold Out)

JSON

```
{
```

```
  "type": "https://api.salestorm.io/errors/out-of-stock",
  "title": "Inventory Unavailable",
  "status": 409,
  "detail": "Requested item is completely sold out for this flash sale event.",
  "instance": "/reservations"
```

- 429 Too Many Requests (Rate Limit Exceeded)

JSON

```
{
  "type": "https://api.salestorm.io/errors/rate-limit-exceeded",
  "title": "Too Many Requests",
  "status": 429,
  "detail": "Rate limit exceeded. Please wait in the virtual queue.",
  "retry_after_seconds": 3
}
```

3.2 Finalize Order (Checkout)

Converts a valid, active reservation into a pending order before payment execution.

- HTTP Method: POST
- Path: /orders/checkout
- Headers: Idempotency-Key: <UUID>

Request Body (application/json)

JSON

```
  "reservation_id": "c3a1b2c3-d4e5-4f6a-8b9c-0d1e2f3a4b5c",
  "shipping_address_id": "addr_99887766"
```

Success Response (201 Created)

```
{
  "status": "SUCCESS",
  "data": {
    "order_id": "ord_8877665544",
    "reservation_id": "c3a1b2c3-d4e5-4f6a-8b9c-0d1e2f3a4b5c",
    "total_amount": 1499.00,
    "currency": "INR",
    "order_status": "PENDING_PAYMENT",
    "created_at": "2026-10-05T12:25:00Z"
  }
}
```

3.3 Initiate Payment Intent

Creates an idempotent payment transaction with Razorpay/Stripe adapters.

- HTTP Method: POST
- Path: /payments/intent
- Headers: Idempotency-Key: <UUID>

Request Body (application/json)

JSON

```
  "order_id": "ord_8877665544",
  "payment_provider": "RAZORPAY",
  "payment_method": "UPI"
}
```

Success Response (200 OK)

JSON

```
{
  "status": "SUCCESS",
  "data": {
    "payment_id": "pay_1122334455",
    "order_id": "ord_8877665544",
    "gateway_client_secret": "rzp_test_sec_998877",
    "amount": 1499.00,
    "currency": "INR",
    "payment_status": "INITIATED"
  }
```

4. OpenAPI 3.0 YAML Definition

Save this complete block as `05_API_Specs/openapi.yaml`:

YAML

```
openapi: 3.0.3
info:
  title: SALESTORM Flash Sale Platform API
  description: High-concurrency API specification for atomic inventory reservation and checkout
processing.
  version: 1.0.0
servers:
  - url: https://api.salestorm.io/v1
    description: Production API Gateway

paths:
  /reservations:
    post:
      summary: Reserve inventory atomically with 10-minute TTL
      operationId: createReservation
      parameters:
        - in: header
          name: Idempotency-Key
          required: true
          schema:
            type: string
            format: uuid
      requestBody:
        required: true
        content:
          application/json:
            schema:
```

```
            type: object
            required:
              - product_id
              - quantity
            properties:
              product_id:
                type: string
                format: uuid
              quantity:
                type: integer
                minimum: 1
                maximum: 2
      responses:
        '201':
          description: Reservation created successfully
        '409':
          description: Out of stock or concurrency conflict
        '429':
          description: Rate limit exceeded

components:
  securitySchemes:
    BearerAuth:
      type: http
      scheme: bearer
      bearerFormat: JWT

security:
  - BearerAuth: []
```